

Academic Appointment

Dec. 2023– **Postdoctoral Fellow**, *Institute for Quantum Computing, University of Waterloo; Perimeter*

Institute for Theoretical Physics, Waterloo, Canada

Education

2017–2023 **Ph.D. in Physics**, *University of Illinois at Urbana-Champaign, Urbana, USA*

2013–2017 **B.Sc. in Physics**, *Nanjing University, Kuang Yaming Honor School, Nanjing, China*

Research Interests

Quantum information and foundations. Entanglement, steering, measurement incompatibility, generalized contextuality, and quantum channels; characterization and certification of quantum resources.

Quantum optics. Entanglement-assisted telescopy, photonic quantum communication, and quantum networks.

AI for Science. AI-assisted research ideation, mathematical exploration, and formal verification in Lean.

Honors and Awards

2023 **Boeing Quantum Creators Prize.** Chicago Quantum Exchange. Recognizes early-career researchers advancing quantum information science and engineering.

Preprints and Publications

5 preprints; 18 peer-reviewed journal articles.

12 first- or co-first-author papers (including 2 co-first-author papers and 2 preprints), with articles in PRX (2), PRL (1), and Quantum (3).

11 additional coauthored papers (including 3 preprints), with articles in PRL (1) and npj Quantum Information (1).

Preprints

[P5] **Quantum key distribution using generalized contextuality against post-quantum eavesdroppers**

D. Centeno, R. D. Baldijão, M. Ciudad Alañón, **Y. Zhang**, P. Lauand, and E. Wolfe.
arXiv:2610.00595 (2026).

- [P4] **Compatibility of quantum instruments**
C. Kim, **Y. Zhang**, E. Chitambar, M. Junge, and F. Leditzky.
arXiv:2609.40266 (2026).
- [P3] **Exact Incompatibility-Breaking Criterion for Unital Qubit Channels**
Y. Zhang.
arXiv:2607.27757 (2026).
- [P2] **Quantum-Limited Subdiffraction Telescopy Requires Genuine Multi-Telescope Interference**
Y. Zhang, Y. Wang, W. Wu, and T. Jennewein.
arXiv:2606.27276 (2026).
- [P1] **Probing the Critical Point (CritPt) of AI Reasoning: a Frontier Physics Research Benchmark**
M. Zhu et al. (including **Y. Zhang**).
arXiv:2509.26574 (2025).

[Peer-Reviewed Journal Articles](#)

- [J18] **Entanglement Certification Using Noncontextuality Inequalities**
Y. Zhang, J. Spodek, D. Schmid, C. Reid, L. J. Morrison, T. Jennewein, K. J. Resch, and R. W. Spekkens.
Phys. Rev. X 16, 031057 (2026).
- [J17] **Quantifiers and witnesses for the nonclassicality of measurements and of states**
Y. Zhang, Y. Yīng, and D. Schmid.
Quantum 10, 2180 (2026).
- [J16] **Reassessing the Boundary between Classical and Nonclassical for Individual Quantum Processes**
Y. Zhang, D. Schmid, Y. Yīng, and R. W. Spekkens.
Phys. Rev. X 16, 021050 (2026).
- [J15] **Time-resolved certification of frequency-bin entanglement over multi-mode channels**
S. Vinet, M. Clementi, M. Bacchi, **Y. Zhang**, M. Giacomini, L. Neal, P. Villoresi, M. Galli, D. Bajoni, and T. Jennewein.
npj Quantum Information 12, 38 (2026).
- [J14] **Criteria for optimal entanglement-assisted long baseline telescopy**
Y. Zhang and T. Jennewein.
Phys. Rev. Research 7, 043278 (2025).
- [J13] **Cost of Simulating Entanglement in Steering Scenarios**
Y. Zhang, J. Zhang, and E. Chitambar.
Quantum 9, 1902 (2025).
- [J12] **Feasibility study of frequency-encoded photonic qubits over a free-space channel**
S. Vinet, W. Wu, **Y. Zhang**, and T. Jennewein.
Opt. Express 33, 40437-40449 (2025).
- [J11] **Temporally Localized Quantum Operations on Continuous-Wave Thermal Light**
Y. Wang, **Y. Zhang**, and V. O. Lorenz.
Phys. Rev. Lett. 135, 113602 (2025).

- [J10] **Astronomical interferometry using continuous-variable quantum teleportation**
Y. Wang, **Y. Zhang**, and V. O. Lorenz.
Phys. Rev. Research 7, 023154 (2025).
 - [J9] **Public quantum network: The first node**
K. Kapoor, S. Hoseini, J. Choi, B. E. Nussbaum, **Y. Zhang** et al.
Appl. Phys. Lett. 126, 054002 (2025).
 - [J8] **Exact Steering Bound for Two-Qubit Werner States**
Y. Zhang and E. Chitambar.
Phys. Rev. Lett. 132, 250201 (2024).
 - [J7] **Information Carried by a Single Particle in Quantum Multiple-Access Channels**
X. Chen*, **Y. Zhang***, A. Winter, V. O. Lorenz, and E. Chitambar.
Phys. Rev. A 109, 062420 (2024).
 - [J6] **Fiber-based photon-pair generation: tutorial**
K. Garay-Palmett, D. B. Kim, **Y. Zhang**, F. Domínguez-Serna, V. O. Lorenz, and A. B. U'Ren.
J. Opt. Soc. Am. B 40, 469-490 (2023).
 - [J5] **Building multiple access channels with a single particle**
Y. Zhang, X. Chen, and E. Chitambar.
Quantum 6, 653 (2022).
 - [J4] **Superresolution in interferometric imaging of strong thermal sources**
Y. Wang, **Y. Zhang**, and V. O. Lorenz.
Phys. Rev. A 104, 022613 (2021).
 - [J3] **Channel activation of CHSH nonlocality**
Y. Zhang*, R. A. Bravo*, V. O. Lorenz, and E. Chitambar.
New J. Phys. 22, 043003 (2020).
 - [J2] **Dual-pump approach to photon-pair generation: demonstration of enhanced characterization and engineering capabilities**
Y. Zhang, R. Spiniolas, K. Shinbrough, B. Fang, O. Cohen, and V. O. Lorenz.
Opt. Express 27, 19050-19061 (2019).
 - [J1] **Adaptive tomography of qubits: purity versus statistical fluctuations**
A. Zhang, **Y. Zhang**, F. Xu, L. Li, and L. Zhang.
J. Phys. A: Math. Theor. 51, 395304 (2018).
- *Equal contribution (co-first authors).

Talks

- 2026 *A paradigm for entanglement certification using noncontextuality inequalities (invited)*. Foxconn.
- 2025 *Criteria for optimal entanglement-enhanced long baseline imaging protocols*. AQIS.
- Defining nonclassicality for individual quantum processes*. QPL.
- 2024 *Exact steering bound for two-qubit Werner states*. QIP.

- 2023 *Quantum-enhanced interferometric imaging (invited)*. Quantum 2.0.
- 2022 *Information carried by a single particle in multiple-access channels*. TQC.
- 2021 *Building multiple-access channels with one particle*. Beyond IID.
- 2020 *Dual-pump design enables novel photon-pair characterization and engineering*. CLEO.
- 2019 *Tailored photon-pair generation in optical fiber through dual-pump spontaneous four-wave mixing*. FiO-LS.

Outreach

- 2023 **Public Quantum Network at the Urbana Free Library**. Led the creation of a public quantum network distributing entangled photons from the University of Illinois to a local library, connecting quantum research with public education.
- 2023 **IQUIST: Ant-Man and the Wasp: Quantumania**. Led post-screening public discussions on quantum science at events coordinated by the Illinois Quantum Information Science and Technology Center (IQUIST).